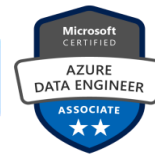


Mamatha M
Senior Data Engineer
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Professional Summary

Senior Data Engineer with **10+ years** of experience building scalable, high-performance data platforms across distributed and cloud environments and expert in designing reliable batch and real-time pipelines, optimizing data models and enabling analytics, reporting and AI-driven insights. Strong at automating workflows, deploying production-grade data systems and translating complex business needs into resilient, cost-efficient data solutions.

- Expertise in handling massive datasets on cloud and distributed platforms including **AWS EMR, Azure Databricks, and GCP Dataproc**, leveraging **HBase, Kafka, Hadoop MapReduce** and **Apache Spark** to optimize large-scale data processing.
- Advanced capabilities in designing and fine-tuning relational databases such as **PostgreSQL** and **SQL Server**, ensuring high-performance queries and reliable transaction management to support Agentic AI applications.
- Demonstrated ability to deliver actionable insights by executing advanced queries with **AWS Athena** and building interactive analytics solutions using **AWS QuickSight, Azure Synapse, Power BI, Tableau** and **Looker**.
- Strong experience in building and managing scalable storage and data warehousing systems across AWS, Azure and GCP, including **RDS, Redshift, DynamoDB, Cosmos DB, Synapse, Snowflake, Cloud SQL** and **BigQuery** to support enterprise-scale data requirements.
- Hands-on experience in implementing reliable database connectivity and application integration using **PyODBC** and **SQLAlchemy**, enabling seamless interaction across multiple systems and heterogeneous data sources.
- Demonstrated experience in designing real-time streaming pipelines using **AWS Kinesis, Firehose, Kafka, Azure Service Bus** and **Azure Stream Analytics** to enable low-latency event processing and near-real-time insights.
- Practical expertise designing and maintaining scalable, **fault-tolerant ETL** pipelines supporting analytics and **Generative AI** workflows, ensuring reliable processing of structured and unstructured data.
- Proficient in deploying applications within containerized environments using **AWS ECS, Fargate, Docker, Kubernetes, Azure Container Instances** and **Google Cloud Run** to ensure flexibility, scalability and simplified deployments.
- Mastery in automating infrastructure provisioning using **Cloud Formation, Terraform, Azure Resource Manager** and **GCP Deployment Manager** to deliver consistent environments, optimized resource utilization and cost-effective scaling.
- Comprehensive experience in designing, training and deploying machine learning solutions using **AWS SageMaker, Azure ML, TensorFlow, Scikit-learn** and **Spark MLlib** to enable predictive analytics and AI-driven insights.
- Familiarity with open-source big data frameworks including **HDFS, Pig, Hive, MapReduce, YARN**, and **Spark** for scalable distributed data processing, analytics, and AI-driven workflow automation.
- Extensive practice in scheduling, orchestrating and monitoring workflows using **Apache Airflow** and **Prefect**, ensuring reliable automation, SLA adherence, optimized performance and production-grade AI/ML pipeline execution.
- Applied experience in streamlining enterprise data integration using **Fivetran** and **Talend**, delivering analytics- and AI-ready datasets with improved data consistency, lineage and reduced migration effort.
- Implemented **IAM, OAuth2.0, VPC**, encryption at rest/in transit, ensuring adherence to **HIPAA, GDPR, and SOX** standards.
- Comprehensive in building interactive dashboards and visualizations using **Looker, QuickSight, Tableau, and Power BI** for enriched datasets into actionable insights, model performance metrics and executive-level decision support.
- Strong capability improving development efficiency and delivery quality through **Git**-based version control and automated **CI/CD** pipelines, enabling continuous deployment of data pipelines, ML feature engineering and AI-enabled services.
- Expert in architecting and managing high-performance data warehouse platforms to support advanced analytics, large-scale reporting, **AI feature stores** and complex business intelligence workloads.
- Proficiency in applying Python extensively for data engineering using libraries such as **Pandas, NumPy, PyArrow, Pytest, Scikit-Learn, Boto3** and **PySpark** to deliver scalable data transformations, feature engineering, model-ready datasets and analytical workflows.
- Practical experience in creating and deploying RESTful APIs using **FastAPI, Flask** and **Django** to expose data services, **AI/ML inference endpoints** and scalable backend integrations for downstream applications.

- Working Familiarity with applying artificial intelligence and natural language processing techniques using **NLTK**, **SpaCy** and **LLM-based prompt engineering** to design intelligent, language-aware systems.

Professional Experience

Client: Zions Bancorporation, Salt Lake City, UT
Senior Data Engineer

Mar 2025 - Till Date

Functional Role Details:

- Engineered real-time financial data pipelines on **AWS EMR** and **Apache Spark**, processing billions of market and transaction events daily to deliver low-latency analytics for regulatory reporting, trading surveillance, AI-driven risk insights, and Agentic AI-powered decision systems.
- Leveraged **AWS Athena** to query petabyte-scale **Amazon S3** financial datasets with sub-second latency, supporting regulatory analysis, trade validation, risk investigations, and LLM-ready structured data access for Generative AI applications.
- Developed interactive financial analytics dashboards using **Power BI** by connecting to **Amazon Redshift** and **curated S3 datasets**, enabling stakeholders to monitor trading activity, fraud indicators, liquidity metrics and regulatory KPIs with near real-time insights.
- Designed and maintained high-performance cloud data warehouses on **AWS Redshift**, optimizing query throughput and storage to support BI dashboards, P&L reporting, ML-based risk feature extraction, and RAG-based financial knowledge retrieval workflows.
- Administered and performance-tuned **PostgreSQL** databases across cloud and hybrid environments, improving availability and query efficiency while enabling low-latency data access for AI models and autonomous agents via API-driven architectures.
- Architected robust streaming solutions using **AWS Kinesis**, enabling real-time fraud detection, liquidity monitoring, and event-driven AI automation frameworks integrating autonomous agents and risk models.
- Automated scalable ETL pipelines with **AWS Glue**, transforming structured and unstructured financial datasets (logs, documents, transaction feeds) into analytics- and LLM-ready embedding formats, improving processing efficiency by 35% and supporting credit risk, AML, and fraud feature engineering.
- Designed and maintained scalable batch and real-time data pipelines to support Agentic AI and Generative AI applications, ensuring reliable ingestion, transformation, and orchestration across financial AI workloads.
- Built and optimized Retrieval-Augmented Generation (RAG) architectures using vector databases and embedding workflows, enabling semantic search and natural-language querying of financial documents and compliance records.
- Processed and transformed large volumes of structured and unstructured financial data (text, logs, PDFs, transaction histories) for LLM training, inference, and AI-powered analytics systems.
- Integrated AI models and autonomous agents with enterprise financial systems via **FastAPI**, event-driven architectures, and streaming frameworks, enabling real-time decision intelligence and AI-assisted operations.
- Implemented data quality validation, monitoring, governance controls, and SLA-based orchestration using **Apache Airflow**, ensuring secure, compliant, and reliable AI-driven financial workflows with 99.9% pipeline reliability.
- Enabled advanced ML workflows using **AWS SageMaker** and **Spark MLlib**, preparing live financial datasets for fraud detection, credit risk scoring, and LLM-enhanced analytics, reducing model deployment timelines by 40%.
- Built and orchestrated containerized real-time financial applications on **AWS Fargate**, delivering serverless elasticity for streaming, AI-enabled workloads, and autonomous automation systems.
- Automated infrastructure provisioning using **AWS CloudFormation**, ensuring secure, compliant, and repeatable environments aligned with financial regulatory standards while accelerating setup times by 70%.
- Applied **DevOps** and **DataOps** practices using **Git** and **CI/CD** pipelines to automate version control, testing, and deployment of data engineering workflows.
- Optimized **AWS EC2**, **Amazon S3**, and networking components to support high-performance AI workloads and automation systems, reducing operating costs by 25% while maintaining low-latency financial pipelines.
- Developed real-time financial data access APIs using **FastAPI** and **SQLAlchemy**, enabling secure, sub-second responses and seamless integration of AI models and autonomous agents with trading and regulatory systems.
- Enhanced financial workflows using LLMs prompt engineering and NLP techniques, automating data transformations, enabling natural-language querying, and generating intelligent insights for operations, compliance, and risk teams.

Functional Role Details:

- Leveraged **scalable streaming pipelines using Google Cloud Dataflow** to automate ingestion from **15+ upstream market and trading systems**, processing **8M+ events daily** and delivering enriched event streams into **BigQuery** enabling **sub-second real-time analytics and operational monitoring**.
- Architected and optimized **Cloud SQL investment data stores** supporting **50M+ trading and portfolio transactions**, implementing **normalized schemas, stored procedures and advanced indexing**, improving **query performance by 40%** and ensuring **low-latency portfolio analytics**.
- Designed executive-ready investment analytics using **Looker**, transforming raw and curated market, portfolio and transaction datasets into intuitive dashboards that enabled real-time, insight-driven investment and risk decisions.
- Implemented event-driven integration patterns using **Pub/Sub** and **Cloud Functions**, enabling real-time triggers, elastic scaling and resilient communication across trading platforms, market data feeds and downstream investment systems.
- Utilized **Google Dataproc and Apache Spark** to process **2TB+ daily market and portfolio datasets**, performing batch analytics, ML feature engineering and complex financial transformations, reducing **processing latency by 35% and compute costs by 25%**.
- Administered and optimized **PostgreSQL on Cloud SQL**, managing provisioning, high availability, backup and recovery and query tuning to ensure consistent performance for trading, portfolio and risk analytics workloads.
- Streamlined investment data ingestion and transformation using **Cloud Data Fusion** and **Dataflow**, improving metadata management, governance and data accessibility across investment data lakes and analytical warehouses.
- Enabled cross-platform interoperability by integrating **Python** investment systems with ODBC-compliant databases via **pyodbc**, simplifying ingestion of trading, pricing and reference data.
- Deployed and operated containerized investment analytics services on **Google Kubernetes Engine (GKE)**, orchestrating **Docker** workloads to deliver scalable, resilient microservices for trading analytics and portfolio services.
- Standardized cloud infrastructure provisioning using **Terraform** and **Deployment Manager**, enforcing infrastructure-as-code practices for consistent, compliant and version-controlled investment platform deployments.
- Strengthened platform security by configuring **IAM, VPC**, subnets and routing policies to ensure controlled access, regulatory compliance and secure communication across investment environments.
- Designed **fault-tolerant data workflows using Cloud Composer (Apache Airflow)** orchestrating **DAG pipelines** for market, pricing and portfolio datasets, improving **pipeline reliability to 99.8% SLA** through automated retries, monitoring and alerting.
- Accelerated development and release cycles through **Cloud Build**, integrating CI/CD pipelines, branching strategies and automated quality checks to support rapid yet stable deployment of investment analytics and trading services.
- Partnered with data science teams to build end-to-end investment ML pipelines using **Vertex AI** and **Scikit-learn**, supporting data preparation, model training and production deployment for pricing, risk and portfolio optimization.
- Developed high-performance investment data utilities using **Python, PyArrow, Pytest** and **Scikit-learn**, enabling efficient data serialization, transformation, validation and automated testing of financial datasets.
- Delivered secure, high-throughput backend investment services using **FastAPI**, exposing RESTful APIs that served portfolio data, market analytics and real-time machine learning inference outputs.
- Integrated **Large Language Models (LLMs)** through prompt engineering techniques to automate data transformation, cleansing and natural-language querying, modernizing analytics and operational workflows.

Functional Role Details:

- Implemented automated ETL orchestration pipelines using **Azure Data Factory**, consolidating data from operational systems, IoT platforms and SCADA sources into centralized energy data lakes and warehouse platforms.
- Collaborated closely with data science teams to productionize machine learning workflows using **TensorFlow**, enabling efficient data preparation, model training and deployment for energy demand forecasting, asset health and predictive maintenance analytics.
- Established secure, high-throughput cloud and hybrid environments by configuring **Azure VMs, Blob Storage, Azure Files** and **VPN Gateway**, supporting controlled connectivity between field systems, control centers and cloud analytics platforms.

- Developed interactive operational analytics dashboards using **Tableau** by connecting curated datasets from **Snowflake** and data lake layers, enabling engineering and operations teams to monitor energy production metrics, equipment health indicators, and predictive maintenance insights across field assets.
- Integrated **Azure Databricks** with **Azure Event Hub** and **Snowflake**, enabling real-time and batch data processing using structured streaming and optimized Spark transformations, reducing end-to-end data processing latency and improving pipeline scalability.
- Architected scalable cloud data warehousing solutions using **Snowflake**, enabling high-performance analytics on energy production, **IoT** telemetry and asset performance datasets while supporting secure data sharing and **BI** integration for enterprise reporting and operational insights.
- Enabled efficient application-to-database integration between **Azure SQL Database** and **Python services** using **SQLAlchemy** and **pyodbc**, enabling scalable, high-performance data access for analytics applications and real-time operational APIs.
- Designed and implemented scalable data processing pipelines using **Azure Databricks (PySpark)**, transforming raw **IoT** and **SCADA** telemetry data into curated bronze, silver, and gold layers, enabling reliable, analytics-ready datasets for downstream reporting and machine learning workloads.
- Built real-time energy data ingestion frameworks using **Azure Event Hub**, enabling continuous streaming of **millions of smart-meter and sensor events per day**, reducing data latency for operational monitoring and predictive analytics by **60%**.
- Developed secure, high-performance backend services using **FastAPI**, exposing APIs that delivered real-time operational datasets and predictive energy insights to both internal platforms and external consumers.
- Containerized data services and API-based microservices using **Docker** and orchestrated deployments on **Azure Kubernetes Service (AKS)**, enabling scalable, fault-tolerant processing environments for real-time energy analytics and **IoT** telemetry workloads.
- Standardized cloud infrastructure deployment using **Azure Resource Manager**, creating reusable infrastructure-as-code templates for consistent, secure and repeatable deployments across environments.
- Managed and tuned enterprise relational database platforms using **Azure Database for MySQL** and **Azure Database for PostgreSQL**, ensuring performance optimization, data integrity and highly available operations.

Client : **AbbVie, Chicago, IL**
Data Engineer

Apr 2019 - Dec 2021

Functional Role Details:

- Deployed scalable serverless container workloads on **AWS Fargate** to support concurrent microservices and data-processing tasks across clinical, manufacturing and pharmacovigilance workflows.
- Built robust end-to-end ingestion and transformation pipelines using **AWS Glue**, consolidating clinical, laboratory and manufacturing data into governed, high-quality data lakes and warehouses.
- Built secure and highly available cloud foundations by configuring **EC2**, **Amazon S3**, **Amazon EFS**, **IAM** and **VPCs**, enabling elastic compute, durable storage and fine-grained access governance aligned with GxP and regulatory requirements.
- Architected high-throughput analytical environments on **Amazon Redshift**, designing optimized schemas, distribution and sort strategies to support large-scale clinical analytics, safety reporting and operational insights.
- Automated cross-system data replication using **AWS DMS** and **Fivetran**, ensuring consistent, near-real-time synchronization of clinical, quality and operational data to support reporting and analytics workflows.
- Designed resilient, event-driven communication using **Amazon SNS** and **Amazon SQS**, enabling loosely coupled, scalable integrations across clinical systems, manufacturing platforms and downstream analytics services.
- Executed distributed data processing workloads on **Amazon EMR**, leveraging **Apache Spark** and **Hadoop** to support large-scale clinical data analysis, ML feature preparation and performance-efficient transformations.
- Accelerated software delivery cycles by integrating **CodePipeline** with **GitHub**, establishing automated CI/CD processes that improved deployment speed, validation and collaboration across regulated development teams.
- Developed high-throughput streaming ingestion pipelines using **Amazon Kinesis**, enabling real-time capture of manufacturing, laboratory and operational events for monitoring, alerting and compliance intelligence.
- Implemented enterprise workflow orchestration using **Apache Airflow** on **Amazon MWAA**, configuring scheduling, dependency management, monitoring and recovery for reliable execution of regulated data pipelines.
- Enforced standardized infrastructure deployment using **AWS CloudFormation**, defining reusable infrastructure-as-code templates for consistent, compliant and repeatable provisioning.

- Operated and optimized relational database platforms on **Amazon RDS**, managing availability, performance tuning and scalability to support both transactional clinical systems and analytical workloads.
- Applied distributed open-source data frameworks including **HDFS, Pig, Hive, MapReduce, YARN** and **Spark** to perform batch processing and enable complex, data-driven analytics across clinical, quality and manufacturing domains.

Client: Ciena Corporation, Hanover, MD
Data Engineer

Aug 2015 - Mar 2019

Functional Role Details:

- Built enterprise-scale pipelines to ingest, transform and load network configuration, topology and telemetry data from relational databases and flat files into centralized repositories for reporting and analytics.
- Optimized batch processing workflows using **SQL** and **Stored Procedures** to transform network performance and log data, ensuring reliable scheduled processing and consistent downstream availability.
- Engineered and tuned **Oracle** and **MySQL** databases, performing schema design, indexing and query optimization to support high-volume network transactional and analytical workloads.
- Implemented distributed batch processing using **Hadoop, MapReduce** and **Hive** to efficiently process large-scale network telemetry and log datasets for analytics and reporting.
- Developed robust network data ingestion and integration workflows using **Informatica, SSIS** and **Talend**, ensuring scalable, reliable and auditable data movement across heterogeneous systems.
- Leveraged **Unix/Linux shell scripting** using **Bash** and **cron** to automate ETL workflows, schedule and monitor batch jobs, implement error handling and recovery and maintain execution logs for operational visibility.
- Created analytics-ready datasets and reporting layers using **Star Schema** and **Snowflake Schema** designs, enabling efficient BI querying, performance-optimized reporting and actionable business intelligence insights.
- Supported reporting and dashboard solutions by delivering curated datasets for tools such as **Cognos, Business Objects, or Crystal Reports**, enabling operational and management insights.
- Ensured data quality and consistency by implementing robust validation checks, reconciliation logic and exception-handling mechanisms across batch data pipelines to maintain accurate and reliable datasets.
- Collaborated with business analysts and application teams to gather data requirements, translate them into technical solutions and deliver reliable, production-ready data assets.

Qualification Badges

- AWS Certified Data Engineer Associate.
- Microsoft Certified Azure Data Engineer Associate.
- Google Cloud Certified Professional Data Engineer.

Educational Chronicle

- Bachelor's in mechanical engineering.